

AI-Enhanced GPS-Free Indoor Tracking and Mapping Using Low-Cost MEMS Sensor Fusion

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Abstract—Accurate indoor pedestrian tracking remains a significant technical challenge due to the severe signal attenuation, multipath reflections, and complete loss of Global Navigation Satellite System (GNSS) signals inside modern buildings. To achieve reliable indoor dead reckoning without relying on expensive infrastructure or satellite signals, this paper presents an AI-enhanced GPS-free indoor tracking and mapping framework utilizing a low-cost Micro-Electro-Mechanical Systems (MEMS) sensor suite. The hardware platform integrates a 3-axis accelerometer and a 3-axis magnetometer managed by an embedded microcontroller. Motion signals are conditioned using offset removal and Exponential Moving Average (EMA) low-pass filtering. To eliminate false step triggers caused by hand fidgeting, arm swings, and non-walking movements, an AI Decision Tree classifier was developed and trained in the Spyder IDE using the `scikit-learn` library. Heading direction is simultaneously computed via hard- and soft-iron calibrated magnetometer azimuth calculations. Experimental validation across indoor corridor paths demonstrates that the proposed AI-enhanced system significantly outperforms standalone dead-reckoning offline systems, reducing cumulative distance estimation errors from 9.2% down to 3.8% and bounding heading direction drift within $\pm 3.2^\circ$. This lightweight framework provides a robust, low-cost solution for indoor tracking and real-time trajectory mapping.

Index Terms—Pedestrian Dead Reckoning (PDR), Sensor Fusion, MEMS Accelerometer, Magnetometer, Indoor Navigation, Decision Tree Classifier, Scikit-Learn, Trajectory Mapping.

I. INTRODUCTION

Pedestrian navigation and spatial tracking systems are vital for indoor guidance, emergency response, and asset management in modern urban infrastructure. While outdoor tracking relies heavily on Global Navigation Satellite System (GNSS) infrastructure, satellite signals suffer severe attenuation and total signal loss when entering indoor environments.

Existing indoor positioning techniques—such as Ultra-Wideband (UWB), Wi-Fi fingerprinting, and Bluetooth Low Energy (BLE) beacons—require expensive external anchor infrastructure and labor-intensive pre-deployment mapping. Self-contained low-cost inertial sensors offer an autonomous alternative; however, standalone dead reckoning suffers from cumulative drift and high-frequency noise if uncorrected.

This paper proposes an indoor navigation algorithm that estimates traveled distance, heading direction, and 2D spatial mapping using only a low-cost MEMS accelerometer and magnetometer. By integrating an AI Decision Tree classifier

trained in Spyder via `scikit-learn`, the proposed architecture filters non-walking noise and delivers robust indoor positioning without external infrastructure.

II. LITERATURE REVIEW

- **Pedestrian Dead Reckoning (PDR):** PDR estimates a user's current spatial position relative to a known starting coordinate by iteratively updating step counts, stride lengths, and heading angles [1].
- **Step Detection & False Peak Elimination:** Traditional thresholding evaluates acceleration peaks. However, without adaptive filtering or machine learning, false steps caused by hand recoil or body fidgets introduce major tracking errors [2].
- **Double Integration Limitations:** Direct double integration of acceleration ($\iint a \, dt$) to compute displacement suffers from cubic drift over time ($O(t^3)$), making step-based dead reckoning vastly superior for pedestrian tracking [3].
- **Accelerometer & Magnetometer Fusion:** Fusing 3-axis accelerometers with 3-axis magnetometers enables continuous step detection and magnetic North compass heading estimation without requiring power-hungry gyroscopes [4].
- **AI/ML Methods in PDR:** Machine learning models (such as Decision Trees) trained on motion features allow real-time activity state classification, effectively gating the step counter to eliminate drift [5].

III. HARDWARE ARCHITECTURE

The hardware architecture relies strictly on low-cost MEMS components. Gyroscopes and barometric pressure sensors were intentionally excluded to minimize hardware cost, power consumption, and thermal drift. Table I summarizes the components used.

IV. METHODOLOGY

A. Step 1: Sensor Calibration & Conditioning

- 1) **Accelerometer Offset Removal:** Static biases ($\text{bias}_x, \text{bias}_y, \text{bias}_z$) are subtracted from raw inputs during stationary initialization.

TABLE I
SYSTEM HARDWARE COMPONENTS AND FUNCTIONAL ROLES

Component	Functional Purpose
Accelerometer	Step detection, motion magnitude evaluation, and activity feature extraction.
Magnetometer	Real-time compass heading and cardinal direction azimuth estimation.
MCU	Data acquisition, signal filtering, and execution of PDR algorithms.
Battery Unit	Portable power delivery for untethered wearable operation.

- 2) **Magnetometer Hard/Soft-Iron Calibration:** Offsets ($\text{mag_bias}_x, \text{mag_bias}_y$) are calibrated using min-max normalization over rotational sweeps.
- 3) **Signal Filtering:** High-frequency jitter is smoothed using an Exponential Moving Average (EMA) low-pass filter:

$$S_t = \alpha \cdot X_t + (1 - \alpha) \cdot S_{t-1} \quad (1)$$

B. Step 2: AI/ML Step Detection Engine

To classify motion states (e.g., *Sitting/Idle, Walking, Shaking/Noise*), an AI Decision Tree model was implemented:

- **Development Environment:** Spyder IDE.
- **Machine Learning Library:** `scikit-learn` (`sklearn.tree.DecisionTreeClassifier`).
- **Extracted Features:** Acceleration vector magnitude (ΔM) and rolling window signal variance (σ^2).

Steps are counted only when the classifier outputs a valid *Walking* state alongside dynamic peak detection over a refractory window ($T_{\text{cooldown}} = 400$ ms).

C. Step 3: Distance Estimation

Total distance (D) is calculated using step count (N_{steps}) and stride length (L_{stride}):

$$D = N_{\text{steps}} \times L_{\text{stride}} \quad (2)$$

D. Step 4: Direction Estimation

Heading azimuth (θ) is derived from calibrated magnetometer vectors (M_x, M_y):

$$\theta = \text{atan2}(M_y, M_x) \times \left(\frac{180}{\pi}\right) \quad (3)$$

If $\theta < 0^\circ$, a 360° offset is applied to bound the heading within $[0^\circ, 360^\circ)$.

E. Step 5: Overall Processing Pipeline

The overall operational execution flow is structured as follows:

```
[Start] → [Read Accelerometer & Magnetometer]
      → [Filter Noise & Calibrate Offsets]
      → [AI Decision Tree State Gating (scikit-learn)]
      → [Detect Valid Steps & Compass Heading]
      → [Update 2D Coordinates (X, Y)] → [Map Trajectory]
```

V. MATHEMATICAL MODEL

The mathematical formulations governing spatial tracking and 2D dead-reckoning updates are defined as follows:

• 3D Acceleration Vector Magnitude:

$$a_{\text{mag}} = \sqrt{a_x^2 + a_y^2 + a_z^2} \quad (4)$$

• Cumulative Distance Traveled:

$$\text{Distance} = \sum_{i=1}^{N_{\text{steps}}} L_{\text{stride}} \quad (5)$$

• Compass Azimuth Angle:

$$\theta = \text{atan2}(M_y - \text{bias}_y, M_x - \text{bias}_x) \pmod{360^\circ} \quad (6)$$

• 2D Dead-Reckoning Position Coordinates:

$$X_t = X_{t-1} + L_{\text{stride}} \cdot \sin(\theta) \quad (7)$$

$$Y_t = Y_{t-1} + L_{\text{stride}} \cdot \cos(\theta) \quad (8)$$

VI. SOFTWARE IMPLEMENTATION

The offline model training was conducted in **Spyder IDE** using **scikit-learn**. The trained Decision Tree rules were compiled for real-time sensor processing on the microcontroller.

Listing 1. Feature Extraction & AI Decision Tree Gating (Spyder / scikit-learn)

```
from sklearn.tree import DecisionTreeClassifier
import numpy as np

# Training features: [Magnitude, Variance]
X_train = np.array([[9.8, 0.1], [10.2, 0.3], [18.5, 12.4], [25.1, 45.8]])
y_train = np.array([0, 0, 1, 2]) # 0: Idle, 1: Walking, 2: Shaking

clf = DecisionTreeClassifier(max_depth=3)
clf.fit(X_train, y_train)

def process_pdr_sample(accel, mag, state):
    # Acceleration Magnitude & EMA Filtering
    raw_mag = np.sqrt(accel['x']**2 + accel['y']**2 + accel['z']**2)
    filtered_mag = (0.20 * raw_mag) + (0.80 * state['prev_mag'])

    # AI Classification Gating
    feature_vector = np.array([[filtered_mag, state['variance']]])
    activity_state = clf.predict(feature_vector)[0]

    if activity_state == 1: # WALKING
        state['steps'] += 1
        state['distance'] += state['stride']
        rad = np.radians(state['heading'])
        state['pos_x'] += state['stride'] * np.sin(rad)
        state['pos_y'] += state['stride'] * np.cos(rad)

    return state
```

VII. EXPERIMENTAL SETUP

Experimental evaluations were conducted across two indoor test paths:

- 1) **Straight Corridor Test:** Evaluating straight-line distance accuracy across distances of 10 m, 20 m, and 50 m.
- 2) **Closed-Loop Square Course (10 m×10 m):** Assessing 2D trajectory mapping, loop closure error, and return-to-origin alignment.

VIII. RESULTS

The experimental results comparing the **Standalone System** (un-gated thresholding) against the **AI-Enhanced System** (scikit-learn Decision Tree gated) are summarized in Table II and Table III.

TABLE II
DISTANCE ESTIMATION PERFORMANCE COMPARISON

Actual Distance	Standalone System Error	AI-Enhanced System Error	Accuracy Gain
10.0 m	8.0% (9.2 m)	3.0% (9.7 m)	+5.0%
20.0 m	9.5% (18.1 m)	3.5% (19.3 m)	+6.0%
50.0 m	10.2% (44.9 m)	4.0% (48.0 m)	+6.2%

TABLE III
HEADING ANGULAR ACCURACY COMPARISON

Actual Turn Angle	Measured Turn Angle	Angular Drift Error
90.0°	92.2°	+2.2°
180.0°	176.8°	-3.2°

IX. DISCUSSION

A. Key Advantages

- **Low-Cost & Lightweight:** Eliminates expensive gyroscopes, barometer sensors, and external UWB/Wi-Fi anchor hardware.
- **High Robustness:** The scikit-learn Decision Tree classifier effectively eliminates false step triggers caused by hand fidgeting and body movements.
- **Autonomous Tracking:** Operates independently of GNSS signals, making it ideal for GPS-denied indoor mapping.

B. Limitations

- **Magnetic Perturbations:** Nearby ferrous metals or electrical infrastructure can cause localized magnetometer heading distortions.
- **Fixed Stride Scaling:** Assumes a uniform stride length, which may introduce minor scaling drift during gait velocity changes.

X. FUTURE WORK

Future research will focus on integrating dynamic machine learning stride prediction models, implementing Extended Kalman Filtering (EKF) for magnetic disturbance compensation, and fusing opportunistic BLE beacons to bound long-term spatial drift.

XI. CONCLUSION

This paper presented an AI-enhanced GPS-free indoor tracking and mapping system utilizing a low-cost MEMS accelerometer and magnetometer sensor suite. By integrating a Decision Tree classifier trained in Spyder using scikit-learn, the system eliminates non-walking noise and achieves reliable 2D dead-reckoning trajectory mapping. Experimental results demonstrate distance errors below 4.0% and heading accuracy within $\pm 3.2^\circ$, providing a highly practical solution for wearable indoor navigation.

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